Solving, Graphing and Displaying Data with GeoGebra

GeoGebra Calculator Suite · GeoGebra

Years 9-10	~30 minutes	Mathematics · Number, Algebra & Statistics

NOTICE — AI-generated worksheet
This worksheet was drafted automatically from the site's curriculum-mapping data and has not been checked by a human curriculum expert. Please review every task for accuracy, safety and suitability for your specific class before use.

Simulation: https://www.geogebra.org/calculator
Curriculum links: AC9M10A01 · AC9M9ST04 · AC9M10N01

Learning intentions

- I can use GeoGebra to solve, expand and simplify algebraic equations and expressions.
- I can choose an appropriate display to represent a data set and justify my choice.
- I can compare exact and rounded (decimal) answers and explain when each is appropriate.

Getting started

Open the GeoGebra Calculator Suite and switch between the Algebra, Graphing and Statistics views as each task tells you to (look for the view icons along the top or side of the screen). Type expressions directly into the input bar — GeoGebra will evaluate, simplify or graph them as you go.

Tasks

1. In the input bar, type an expression such as 3(x+4)-2x and press enter. Record what GeoGebra shows as the simplified result.

2. Now type an equation such as 2x+5=17 and enter it. What does GeoGebra return, and how is this different from what happened with the expression in task 1?

3. Choose a linear equation of your own (for example, one from a recent homework question) and enter it into GeoGebra. Show the equation you used and the solution GeoGebra gives you.

Working:

4. Enter three different fractions (e.g. 1/2, 2/3, 5/6). For each, record the fraction, the decimal form GeoGebra displays, and whether that decimal is exact or rounded.

Fraction	Decimal form shown	Exact or rounded?

5. Using your table, explain in your own words why the exact and decimal answers for the same fraction are not always identical.

6. Open the Statistics or Spreadsheet view and enter this set of test scores: 12, 15, 15, 18, 20, 24. Use GeoGebra's tools to produce a bar chart and a box plot of this data.

7. Compare the bar chart and box plot you produced. Which display would you choose to show a friend how spread out the scores are, and why?

8. Use GeoGebra to calculate the area of a rectangle with side lengths 4.7 cm and 3.2 cm, first using the exact decimals, then again after rounding each side to 1 significant figure before multiplying. Record both answers.

Working:

Extension (optional)

A bank statement rounds every transaction to the nearest cent. Use GeoGebra to investigate what happens if you repeat a rounded calculation (like compound interest) many times over — does the rounding error grow, shrink, or stay about the same? Explain your reasoning.